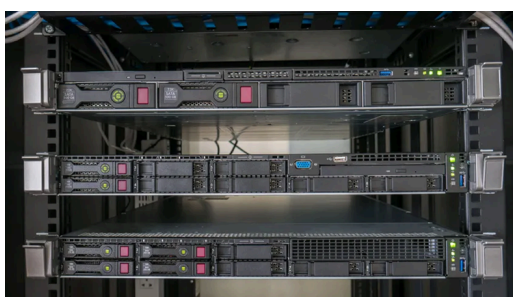


Types of Servers



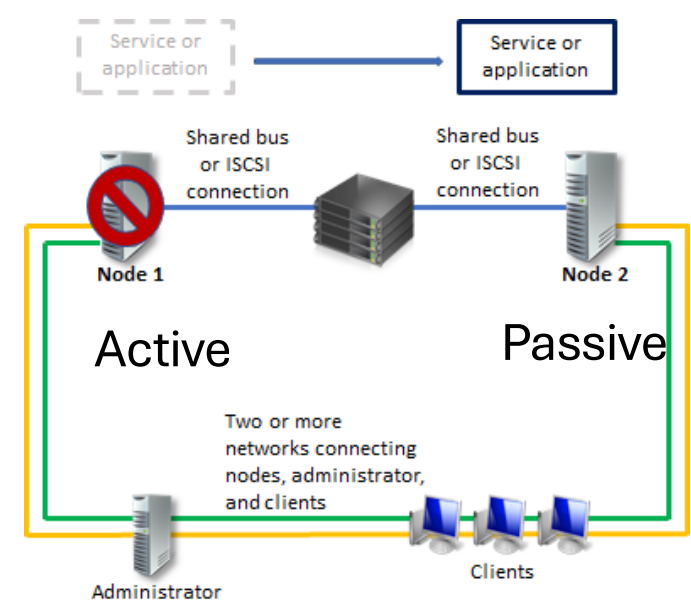
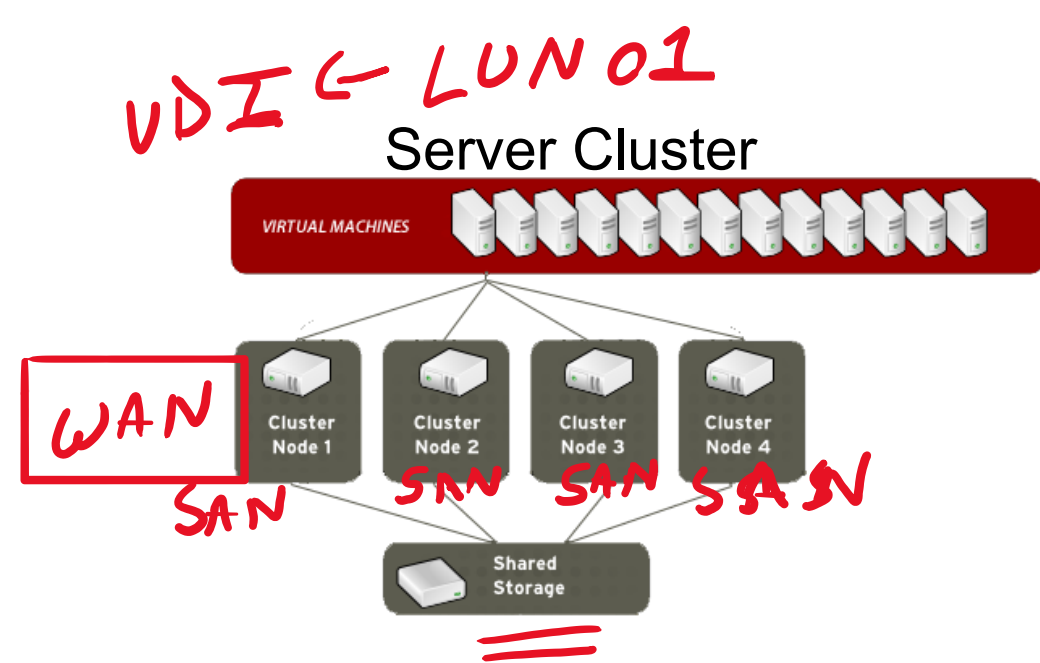
Tower



Rack



Blades



Failover Cluster

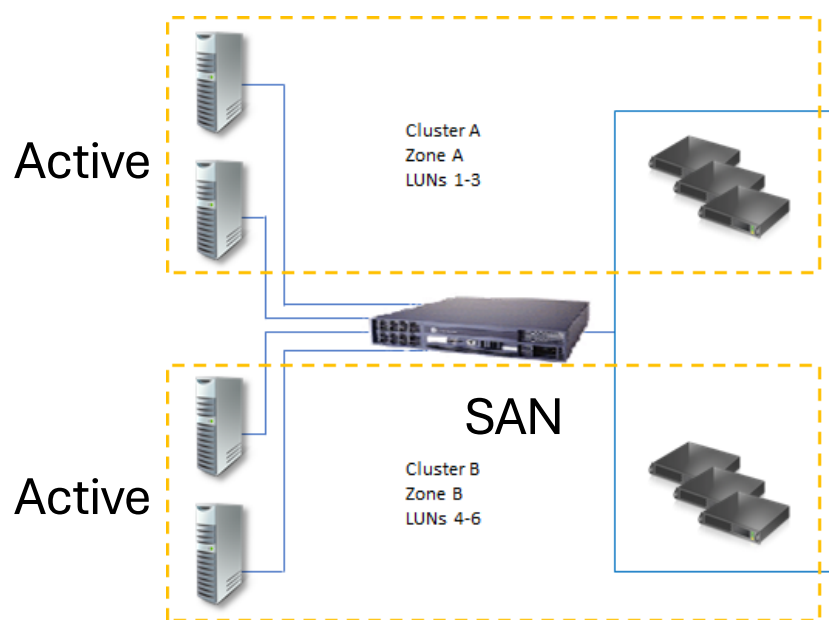
1. Cost-effective: Generally more affordable for small to medium-sized businesses (SMBs) than rack or blade servers where dedicated server room is not necessary but highly recommended.
2. Standalone, Vertical Design: Tower servers have a standalone, vertical chassis resembling a desktop PC.
3. Occupies more physical space compared to rack-mounted servers not suitable for datacenter.
4. More expansion slots to scale Hardware Resources within a Chassis support scaling of hardware resources.
5. For cooling solution, they are built with multiple fans that support efficient cooling.
6. Good for dedicated network services for stable service deployment and delivery in Large Enterprises, services include file storage, application hosting and local services.
7. Built-in Power Supply Unit, to support hardware power requirements.

1. Space requirement - full rack can hold up to 20 to 42 servers, depending on number of servers and additional storages.
2. Physical space - 19-inch width racks, common sizes 1U, 2U, 4U. Measured in rack units (U), commonly 1U to 4U tall. Height 1 U= 1.75 " inches.
3. Built with Swapable Hard drives in front.
4. They have small fan built in which makes them noisy compared to Tower.
5. Deployment - deployed in rows to optimise airflow and cooling, use cable management for airflow and accessibility.
6. Cable management is necessary to support with better Air flow and overall network management.
7. UPS and invertors can be added to the rack to provide uninterrupted power supply.
8. Power consumption is higher compared to Tower.
9. Noisier than tower servers as they have smaller but higher RPM fans.
- 10 . Best for deployment in Medium to Larger scale or datacenter businesses.



1. Most compact of the three. It uses chassis to house multiple blades.
2. Space efficient with multiple server with Vertical blades modules in one chassis.
3. High-performance computing, best for high-density setups.
4. Ideal for large-scale enterprise data centers.
5. Storage Solution, Identity management, Large Language Model, VDI Virtualisation.
6. Shared Power supply Unit and network ports for a single chassis.
7. Cable Management is better with less cables used compared to Rack.
8. Louder than other Servers

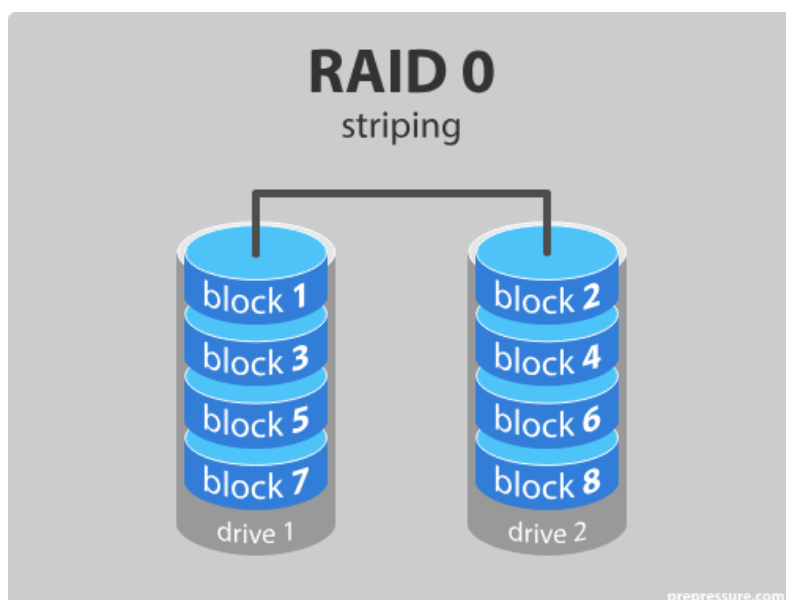
1. High Availability, Load balancing, Fault Tolerant, Redundancy, Scalability , Centralised management and Fail-Over.
2. Performance and multi point failure.
3. Higher the availability higher the Cost
4. Need dedicated cooling solution to support function.
5. High maintenance
6. Works through shared resources.
7. Business critical application such Big Data to minimise the downtime and support service availability.
8. Cluster Threshold decides the status of the cluster to active or passive.
9. Single quorum server cluster help manage cluster load for service request.
10. Needs Cluster management tools to manage cluster servers and their services.



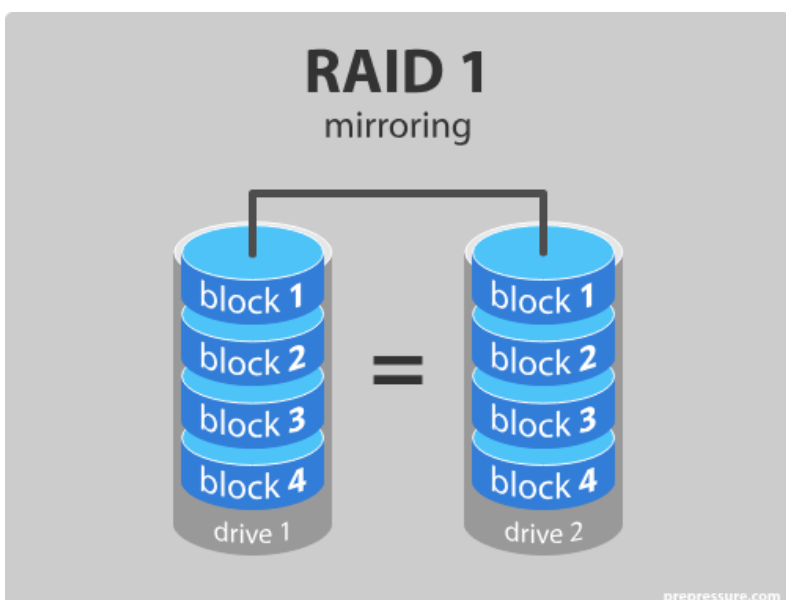
Storage 1

Storage 2

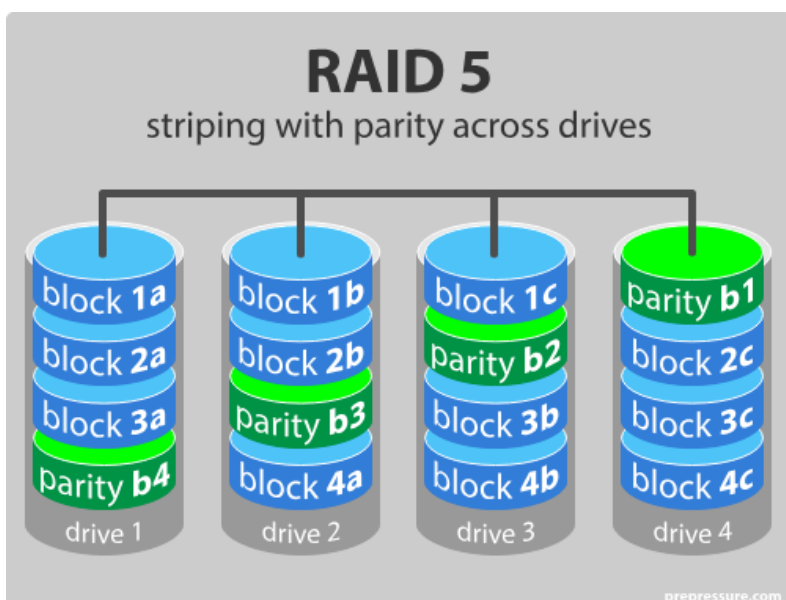
Storage Solutions



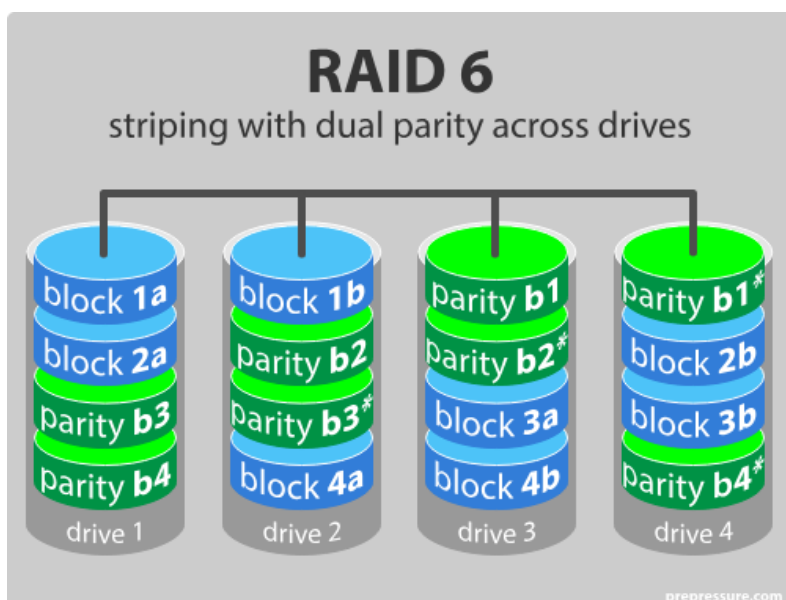
RAID 0 – Striping



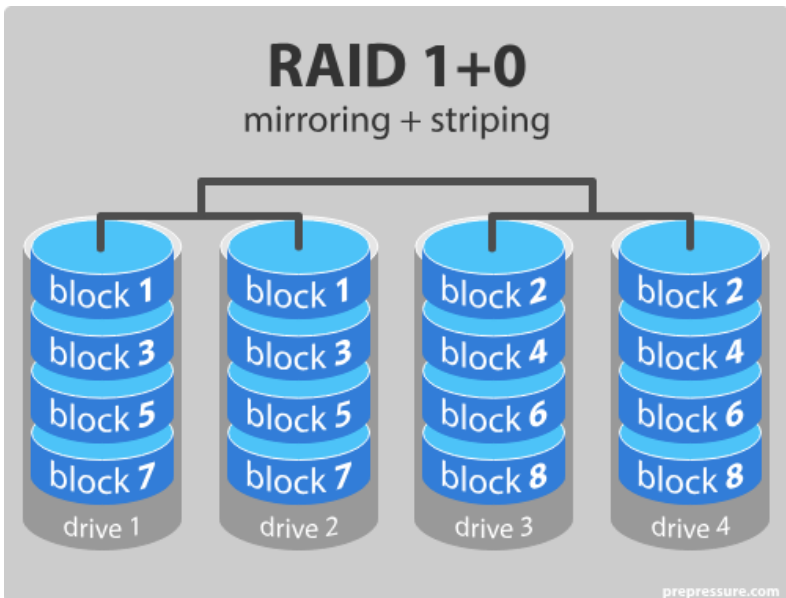
RAID 1 – Mirroring



RAID 5 – Striping with parity



RAID 6 – Striping with double parity



RAID 10 – Striping and mirroring